Chapter 15
Particle-in-Cell and Hybrid Simulation

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Abstract The aim of this lecture is to provide a very short 'primer' course on the PIC method. This will cover the basics of PIC algorithms and numerical methods, potential pit-falls of the method, and the important extensions to the method (including so-called 'hybrid' codes). This lecture is intended to be a starting point for further study, however enough details are given for a student to write their own 1D PIC code with some extra work.

15.1 The Kinetic Equation

The mathematical foundation of the kinetic approach is found in the kinetic equation for the distribution function, $f = f(\mathbf{r}, \mathbf{p}, t)$ of a given species,

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{\gamma m}\nabla_{\mathbf{r}} f + q(\mathbf{E} + \frac{\mathbf{p}\times\mathbf{B}}{\gamma m}\nabla_{\mathbf{p}} f = C(f) + S(f). \tag{15.1}$$

If the left hand side is equated to zero then this becomes the well known *Vlasov* equation which models a collisionless plasma. The right hand side contains the collision operator, $C(f)$, and potentially a source term, $S(f)$ if ionisation/recombination processes are present. The single-particle distribution function, f, is valid if there are no strong correlations between individual particles, which is valid in weakly-coupled plasmas.

The collision operator can be specified mathematically via the Fokker-Planck equation, although since we are initially concerned with fully ionised, collisionless plasmas we will not consider either $C(f)$ or $S(f)$ in much detail. Importantly, one

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will need to couple the kinetic equation to Maxwell's equations, noting that only two are actually required,

$$\frac{\partial \mathbf{E}}{\partial t} = -\frac{\mathbf{j}}{\varepsilon_0} + c^2 \nabla \times \mathbf{B}, \tag{15.2}$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}. \tag{15.3}$$

The source term (current density), $\mathbf{j}$, can be obtained from the first moment of the distribution function. One now has a closed set of equations.

The problem with using this directly, i.e. by gridding f and finite differencing the Vlasov equation, as a basis for numerical simulation is that this rapidly becomes very computationally demanding. The distribution function at any given time can be up to six dimensional in nature. If N grid points are used for each dimension, then the storage requirement will scale as N^6. For storing each grid point as an eight byte double precision number this means that if $N = 100$ one would require over 7,000 GB to store this data structure. For lower dimensional systems (e.g. $f(x, p_x)$) the requirements are feasible, and indeed finite difference Vlasov simulation for those systems are possible. Nonetheless a reduction of the computational overheads is required.

15.2 The Particle Approach

One way to achieve 'information reduction' is to note that any distribution function can be approximated by a set of 'clouds' or 'macroparticles' that have some finite volume in phase space. Suppose we were to track a finite set of such macroparticles that each had a fixed form. What equations would have to be solved? Let,

$$f = \sum_i g_i(\mathbf{r} - \mathbf{r}_i(t), \mathbf{p} - \mathbf{p}_i(t)). \tag{15.4}$$

If we substitute this into the Vlasov equation and resolve the time derivative by using the chain rule then we quickly find that,

$$\frac{d\mathbf{r_i}}{dt} = \frac{\mathbf{p_i}}{\gamma_i m_i}, \tag{15.5}$$

and

$$\frac{d\mathbf{p_i}}{dt} = \frac{q_i (\mathbf{E} + \mathbf{v} \times \mathbf{B})}{\gamma_i m_i}. \tag{15.6}$$

In other words, we find that the 'macroparticles' can be evolved as a set of single particles. Although this is intuitively obvious, it shows that particle simulation is a

fully consistent approximation to the Vlasov-Maxwell system. This approach has other advantages as well. A finite difference approach allows the possibility that a poor algorithm will drive f negative in some region of phase space, something which is impossible in this approach. One also notes that although a Vlasov code may have to handle regions of phase space that are empty, a PIC code does not, and it is therefore more efficient in this sense. One therefore expects this to produce highly 'robust' codes. On the other hand, this then raises questions about how accurate this method will be for a given number of macroparticles, and whether it leads to unphysical behaviour or phenomena. Since we will still solve Maxwell's equations on a grid that the macroparticles move through, we will have to interpolate onto the grid to obtain current densities and interpolate from the grid to obtain the electromagnetic fields at each macroparticle. This interpolation has the potential to cause a number of non-physical effects. These matters will be considered in more detail later on.

The central PIC algorithm will therefore proceed as follows:

1. Move the macroparticles.
2. Interpolate onto grid to obtain current densities.
3. Update the electromagnetic fields.
4. Interpolate from the grid to obtain EM fields at each macroparticle.
5. Update the momentum of the macroparticles.

Clearly choices must be made about the overall integration scheme. One common choice is to store the particle positions at $t_n = n\delta t$, and the particle momenta at $t_{n+1/2} = (n+1/2)\delta t$,i.e. the 'Leapfrog' scheme. Once these details are determined we can deal with each of these key steps in turn. In what follows we will describe things primarily in terms of a 1D3P, relativistic and electromagnetic code, but we will add important details that are relevant to 2D and 3D algorithms where appropriate.

15.3 Particle Pusher

The position and momentum updates constitute what might be termed the "particle pusher" part of the algorithm. The simplest step is the update of the macroparticle positions. Assuming that we have adopted a Leapfrog scheme, the step that must be performed in the 1D3P algorithm on each macroparticle at the $(n+1)$-th time-step is,

$$\frac{x^{n+1} - x^n}{\Delta t} = \frac{p_x^{n+1/2}}{\gamma^{n+1/2} m}. \tag{15.7}$$

The momentum update is only slightly more complex, and the complicating factor is the $\mathbf{v} \times \mathbf{B}$ term in the Lorentz force. One algorithm that does this effectively is the *Boris* algorithm, which separates the electric field and the $\mathbf{v} \times \mathbf{B}$ rotation. There

are three main stages to this. Firstly one accelerates the macroparticle in the electric field for half a time step,

$$\mathbf{p_i^-} = \mathbf{p_i^{n-1/2}} + \frac{1}{2} q \mathbf{E^n} \delta t. \tag{15.8}$$

Next the $\mathbf{v} \times \mathbf{B}$ rotation is performed. Since this does no work on the macroparticle, γ is constant throughout this step, and we denote this as γ^*. The difference equation that is solved, for $\mathbf{p}_i^+$, is,

$$\frac{\mathbf{p}^+ - \mathbf{p}^-}{\Delta t} = \frac{q}{2\gamma^* m} \left(\mathbf{p}^+ + \mathbf{p}^- \right) \times \mathbf{B^n}. \tag{15.9}$$

This can be re-arranged into a set of three coupled equations (assuming 3P) for $\mathbf{p}^+$, which can be cast in matrix form and solved by inverting this matrix.

Finally, the macroparticle is again accelerated by the electric field for half a time step.

$$\mathbf{p^{n+1/2}} = \mathbf{p}^+ + \frac{1}{2} q \mathbf{E^n} \delta t. \tag{15.10}$$

15.4 Interpolating to and from the Grid

The interpolation between particles and the grid and vice versa (Particle-Grid Interpolation; PGI), depends on what choice is made for the spatial form of the macroparticles, or what is also called the *shape factor*, $S(\mathbf{r})$. In 1D, the i-th macroparticle will have an associated number of (real) particles per unit area, $\mathcal{N}_i$, and charge per unit area σ_i. The particle number and charge density in the j-th spatial cell is then given by,

$$n_j = \sum_i \mathcal{N}_i S(x_j - x_i)/\Delta x, \tag{15.11}$$

and,

$$\rho_j = \sum_i \sigma_i S(x_j - x_i)/\Delta x. \tag{15.12}$$

Other quantities such as current density can be obtained in the same way. In order to interpolate, for example, the E_x component of the electric field from the grid onto the particle in 1D, one has,

$$E_{x,i} = \sum_j E_j S(x_j - x_i). \tag{15.13}$$

One cannot proceed without making a definite choice for $S(x)$. Choosing a very simple top hat function of width Δx,i.e. *Nearest Grid Point* weighting, will result in simple coding whereas using linear interpolation or quadratic splines

will require somewhat more coding. On the other hand higher order interpolation improves accuracy, smoothness, and other aspects (including energy conservation). To illustrate this, consider the case of linear interpolation in 1D, each macroparticle will contribute to only two grid points and vice versa, i.e.

$$\rho_j = \frac{\sigma_i}{\Delta x}\frac{x_{j+1}-x_i}{\Delta x}, \tag{15.14}$$

and,

$$\rho_{j+1} = \frac{\sigma_i}{\Delta x}\frac{x_i-x_j}{\Delta x}. \tag{15.15}$$

For the electric field we would then have contributions from only two grid points,

$$E_{x,i} = E_{x,j}\frac{x_{j+1}-x_i}{\Delta x} + E_{x,j+1}\frac{x_i-x_j}{\Delta x}. \tag{15.16}$$

PGI in 2D and 3D obviously involves more work, but is not conceptually different. PGI is a very critical part of the PIC algorithm in the sense that it is the point where the majority of non-physical behaviour can emerge. Firstly, PGI opens the possibility of a macroparticle exerting a force on itself. If, however, one ensures that the same $S(x)$ is used in both interpolation steps then this self-force can be eliminated. Secondly, using the same $S(x)$ throughout will ensure conservation of momentum (at least with periodic boundary conditions). It is possible to devise energy conserving algorithms, but these will then not conserve momentum exactly.

15.5 Solving Maxwell's Equations

Once PGI has taken place, one now has the source terms required for advancing the electromagnetic fields by one time step. The discussion here starts with the general approach used in 2D and 3D. The standard approach is to define the fields on the grid not at cell centres but with the electric fields along the cell edges and the magnetic fields normal to the cell faces as was proposed by Yee. Once this is done, one can difference Maxwell's equations in a way that is second order accurate in time and space. Here this is illustrated for the case of (E_x, E_y, B_z) in 2D. The scheme is a leapfrog method in essence, with the electric field stored at $n\Delta t$ and the magnetic flux densities and current densities at $(n+1/2)\Delta t$. The induction equation becomes,

$$\frac{B^{n+1/2}_{z,i,j} - B^{n-1/2}_{z,i,j}}{\Delta t} = -\frac{E^n_{y,i+1,j+1/2} - E^n_{y,i,j+1/2}}{\Delta x} + \frac{E^n_{x,i+1/2,j+1} - E^n_{x,i+1/2,j}}{\Delta y}, \tag{15.17}$$

and the electric field equations become,

$$\frac{E^{n+1}_{x,i+1/2,j} - E^n_{x,i+1/2,j}}{\Delta t} = -\frac{j_{x,i+1/2,j}}{\varepsilon_0} + c^2\frac{B^{n+1/2}_{z,i+1/2,j+1/2} - B^{n+1/2}_{z,i+1/2,j-1/2}}{\Delta y}, \tag{15.18}$$

and,

$$\frac{E^{n+1}_{y,i,j+1/2} - E^{n}_{y,i,j+1/2}}{\Delta t} = -\frac{j_{y,i,j+1/2}}{\varepsilon_0} + c^2 \frac{B^{n+1/2}_{z,i+1/2,j+1/2} - B^{n+1/2}_{z,i-1/2,j+1/2}}{\Delta x}. \quad (15.19)$$

One issue that requires care, is that of ensuring that $\nabla.\mathbf{B} = 0$ and $\nabla.\mathbf{E} = \rho/\varepsilon_0$. Provided that $\nabla.\mathbf{B} = 0$ initially, this difference scheme will ensure that it is maintained. Ensuring that Gauss' Law is upheld requires that the $\mathbf{j}$ and ρ used satisfy the continuity equation. This particular problem bears some similarities to ensuring that $\nabla.\mathbf{B} = 0$ in MHD codes. One solution is to employ *divergence cleaning*, i.e. an extra solve is carried out, but divergence cleaning is not a particularly satisfactory solution. The problem can be eliminated by computing $\mathbf{j}$ in a fully charge conserving fashion (first done by Buneman/Morse and Nielson), although this has implications for the noise properties of the EM fields.

In a 1D PIC code some simplifications can be made. The electric field component that coincides with the single spatial direction (which we assume to be E_x) can be determined by integrating Gauss' Law,

$$\frac{\partial E_x}{\partial x} = \frac{Zen_i(x) - en_e(x)}{\varepsilon_0}. \quad (15.20)$$

The transverse field components can then be updated by noting that, for the $E_y - B_z$ case, if we define,

$$F^{\pm} = \frac{E_y \pm B_z}{2}, \quad (15.21)$$

then $F^{\pm}$ are advected at c in the right/left directions respectively with a source term of $-j_y/2$. Therefore if the time step is set to $\Delta x/c$ then the update can be performed via,

$$F^{\pm,n+1}_{j} = F^{\pm,n}_{j\mp1} - \frac{\Delta t}{4}\left(j^{-}_{y,j\mp1} + j^{+}_{y,j}\right), \quad (15.22)$$

where the $-$ superscript on j_y denotes the current density obtained by interpolating the particles onto the grid at t, and the $+$ superscript denotes the current density obtained by interpolating at $t + \Delta t$. This ensures that the current density is centred in space and time.

15.6 Boundary Conditions and Lasers

In the previous sections we have covered all of the core components of a PIC code, but have thus far omitted any discussion of how the boundaries of the computational domain are handled. As PIC codes are grid-based, and the grids must be finite in extent, deciding on appropriate boundary conditions is unavoidable. There are two issues to be decided: (i) What happens to a macroparticle on reaching the grid boundary?, and (ii) What are the boundary conditions on the EM fields?

The typical choices that are made for handling macroparticles at the boundaries include:

- Periodic: Macroparticles that leave through one boundary, reappear at the other with unchanged momentum.
- Reflective: Macroparticles undergo specular reflection at boundaries.
- Open/Absorbing: Macroparticles are lost on passing across boundaries.

Each choice then has implications for the boundary conditions on the EM fields, e.g. if the boundaries are periodic for the macroparticles then the same should apply for the EM fields. In 1D codes the domains can often be made so large on modern computers that they will not affect the system by any appreciable amount. Periodic boundary conditions are easy to implement. So are boundaries that are reflective for the particles. In this case, $E_x = 0$ at these reflective boundaries.

In 2D PIC codes, the two boundary conditions that are used most often are periodic and open boundaries. In many studies, the boundaries are periodic in the direction transverse to the laser and open in the direction parallel to the laser. This arrangement is useful as long as no perturbations reach the transverse boundaries (and propagate back to the centre), as it is often used to simulate systems that have no real periodicity in the transverse direction at all. In many problems, however, it is hard to avoid scattered electromagnetic waves reaching the transverse boundary with a reasonably sized computational domain. In computational electromagnetics, a solution to this problem is to use 'Perfectly Matched Layers' at the transverse boundaries which will absorb these EM waves rather than scatter them back into the box. Perfectly Matched Layers have been implemented in some codes.

Laser pulses can be incorporated into PIC codes in two ways. If there is a large enough vacuum region available, a laser pulse could, in principle, simply be "loaded" onto the grid as an initial condition. In 2D and 3D, this is not an 'affordable' solution so laser pulses are instead launched into the grid at the boundaries.

15.7 Initialisation

The final physics-based core component of a PIC code is the algorithm used to initialise the macroparticles. Laser-plasma researchers are often interested in the relatively simple case of a neutral, field-free, stationary plasma. It is straightforward to assign the correct weight to each macroparticle to ensure that the correct particle density is achieved in each spatial cell. What is slightly non-standard is the initialisation of the distribution function in momentum. Any distribution function can be obtained from a set of uniformly distributed random numbers by the inversion of the cumulative distribution function, i.e. calculate,

$$D(\mathbf{p}) = \frac{\int_0^{\mathbf{p}} f(\mathbf{p}) d^3\mathbf{p}}{\int_0^{\infty} f(\mathbf{p}) d^3\mathbf{p}}, \tag{15.23}$$

then by equating $D(\mathbf{p})$ to a set of uniformly distributed random numbers, one will obtain a set of **p**s that follow the specified distribution function.

This method of initialising the macroparticles has sometimes been found to be undesirable on the grounds that it is highly noisy. An alternative is to initially distribute the macroparticles in a proscribed order, which is completely without the use of random numbers. This is sometimes referred to as a 'quiet start'.

15.8 Key Issues with PIC Codes

It should be apparent by now that the PIC method is not without it potential pit-falls. This is not to say that the method is fundamentally flawed — it is very powerful — but to point out that the method is prone to error if abused. The first point to make concerns the number of macroparticles, and especially the number per cell. The number of macroparticles per cell (N_{pc}) determines how well the distribution function can be represented in phase space. Clearly one macroparticle in a cell can represent no more than a cell in a hydrocode, but beyond this it rapidly becomes difficult to determine *a priori* whether a given number of particles per cell will be sufficient for a given problem. Often this problem needs to be addressed in an 'experimental' manner, in order to determine whether or not convergence has been attained. Related to this is the problem of noise in PIC codes. PIC codes suffer from shot noise in the gridded quantities that scales as $\propto 1/\sqrt{N_{pc}}$. This is not physical and can often obscure physical phenomena that one is trying to observe.

Even if one has a reasonably high N_{pc} one is not guaranteed to be free of non-physical behaviour. PIC codes can exhibit non-physical instabilities, particularly when the Debye length is shorter than the spatial cell size and the time step is greater than the plasma period. These problems have been examined very thoroughly using formal mathematical theories. Here we will by-pass the formal theory and present results where possible. Firstly, we will look at the leap-frog integrator and how it handles the case where a single macroparticle becomes a simple harmonic oscillator. The difference equations that span two time steps from $t - \Delta t$ to $t + \Delta t$ are:

$$\frac{x^{n+1} - x^n}{\Delta t} = v^{n+1/2}, \tag{15.24}$$

$$\frac{v^{n+1/2} - v^{n-1/2}}{\Delta t} = -\omega_0^2 x^n, \tag{15.25}$$

$$\frac{x^n - x^{n-1}}{\Delta t} = v^{n-1/2}. \tag{15.26}$$

These three equations can be combined into a single equation for the xs alone. If one now looks for solutions of the form $x^n = Ae^{i\omega t}$, then one obtains the following expression,

$$\sin\left(\frac{\omega t}{2}\right) = \pm\frac{\omega_0 \Delta t}{2}. \tag{15.27}$$

This immediately tells us numerical instability can occur if $\omega_0 \Delta t > 2$, and we will only obtain good accuracy (i.e. $\omega \approx \omega_0$ if $\omega_0 \Delta t/2 \ll 1$. This tells us that the highest natural frequency of the system, i.e. the plasma frequency, needs to be well resolved in time. In interactions with underdense plasmas, where the laser frequency becomes the highest frequency of the system, the same consideration applies.

Secondly we will look at the need to resolve the Debye length. This is somewhat more subtle, as it involves non-physical modes (*aliases*) which can destabilise the plasma. For 1D warm plasmas, the electrostatic dielectric functions can be derived for both the physical case and the PIC case. One finds that the two will only coincide if $\Delta x \ll \lambda_D$. If one doesn't start a PIC simulation of a warm plasma with the Debye length resolved then non-physical electrostatic modes will artificially heat the plasma until the Debye length is resolved. Using smoother functions for macroparticle spatial profile can mitigate this *numerical heating* but not stop it altogether. Note that numerical heating is consistent with our previous comments on energy-momentum conservation in PIC codes, i.e. momentum-conserving PIC algorithms do not conserve energy exactly.

We shall conclude this section by saying the following: PIC codes are excellent tools for understanding real physics, but one must be aware of their limitations and take great care not to abuse them!

15.9 Extending the PIC Method

In the previous sections the reader will have seen that, despite certain limitations, PIC codes are robust and based on relatively simple algorithms. Their simplicity and robustness has lead a number of researchers to try to extend the method to much more than just fully ionised, collisionless plasmas. Hybrid PIC codes are covered in the next section, so here we will discuss a number of other developments of the PIC method:

- **Collisions**: Collisions can be included through Monte-Carlo algorithms. Such methods normally involve randomly pairing macroparticles within each spatial cell, and performing binary Coulomb collisions. It can be shown that, in the limit of large N_{pc}, this is equivalent to the Fokker-Planck collision operator. The additional operations do increase the run-time. Whether or not physically correct transport coefficients are reproduced for a viable N_{pc} needs to be checked.
- **Ionisation**: With an appropriate knowledge of ionisation cross-sections, one can also use Monte-Carlo methods to include electron and ion impact ionisation processes. Field ionisation can similarly be included. How well momentum space is represented by a given N_{pc} is equally important here too.
- **Radiation Reaction**: When charged particles are accelerated they emit radiation. If this acceleration becomes extremely strong then the radiation emission exerts

a non-negligible back-reaction on the charged particle. It is thought that this will become important at the extreme intensities that will be realised by future laser systems ($\approx 10^{23}\mathrm{Wcm}^{-2}$).

- **High Energy Photons and QED processes**: At extreme laser intensities ($\gg 10^{23}\mathrm{Wcm}^{-2}$) it is possible that very high energy photons (generated by violent acceleration of electrons) will disintegrate into electron-positron pairs in the ultra-strong EM fields. Current research aims to include both photon and pair production into PIC codes via Monte Carlo methods.

15.10 Hybrid Codes

There is considerable interest in subjects such as fast electron transport, and fast ignition ICF, where a small population of super-thermal (a.k.a. *fast*) electrons with very long mean free paths propagate through a dense, collisional, and relatively cool plasma. This is a very demanding problem due to the great disparity in length and time scales. Collisionless PIC codes cannot model the resistivity of the background plasma, and thus can't capture effects due to the finite resistivity of the background plasma (especially magnetic field generation).

One model that has been used to study fast electron transport is the *hybrid code*. The term 'hybrid code' is used in a number of different areas of plasma physics with different meanings. In this context it means that the background plasma is treated as a fluid, and the fast electrons are treated kinetically. Hybrid codes make a number of assumptions that one needs to be aware of:

- **Small Fast Population**: Most importantly, hybrid codes make the assumption that the fast population is much smaller than the background population everywhere, i.e. $n_f \ll n_b$, even though the current density of the fast need not be negligible. Without this, it would be impossible to treat the background as a distinct, quasi-neutral plasma.
- **Validity of Fluid Background**: Hybrid codes also assume that, even if the fasts were absent, that a fluid description of the background plasma is valid on the time and space scales of interest. Since a number of properties of the fluid background will be prescribed, e.g. resistivity, specific heat capacity, etc., all of these models need to be valid and accurate as well.
- **Current Balance**: Provided that the fast electrons are a small population there are strong electrostatic and magnetostatic arguments for current balance holding, i.e.

$$\mathbf{j}_f + \mathbf{j}_b = \frac{\nabla \times \mathbf{B}}{\mu_0}, \tag{15.28}$$

where the 'f' subscript denotes the fast electrons and the 'b' subscript denotes the background electrons. Current balance will only break down if j_f varies strongly

over the background electron's collisionless skin depth. If the fast population is indeed very small then this should not occur. The displacement current is also neglected.

- **Ohm's Law**: Hybrid codes assume an Ohm's law to determine the electric field. The simplest one that is typically used being,

$$\mathbf{E} = \eta \mathbf{j}_b, \tag{15.29}$$

which, on assuming current balance and neglecting $\nabla \times \mathbf{B}$ becomes,

$$\mathbf{E} = -\eta \mathbf{j}_f. \tag{15.30}$$

- **Resistive Magnetic Field Generation**: Although not an assumption *per se*, the use of Ohm's Law to determine the electric field, must reduce the induction equation. For the simple Ohm's law this will result in,

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times \left(\eta \mathbf{j}_f \right), \tag{15.31}$$

which expresses the resistive generation of magnetic field around a fast electron beam.

These assumptions greatly relax the resolution requirements for these problems, since the cold and dense background has very small spatial and temporal scales that make fully kinetic simulation unfeasible. They also greatly simplify the algorithm, as the reduced Maxwell's equations as much easier to solve. Fast electrons are simply introduced onto the grid in a 'heating zone' according to some prescription that models fast electron generation by the laser interaction. The fast electron macroparticles will also undergo scattering and drag by the background via Monte-Carlo algorithms. Apart from these modifications the same PIC algorithms are employed.

15.11 Summary

At the most basic level, the PIC method is simple and intuitive. The numerical methods that it employs are fairly transparent, as well as being straightforward to understand and program. This can be deceptive however. PIC codes are also subject to a range of non-physical behaviour, which includes statistical noise, non-physical instabilities, non-conservation, and numerical heating. One simply cannot expect to achieve good quality kinetic simulation by simple means without 'paying the piper'!! Having said this, with due care and attention they are excellent numerical tools, that can provide great insight into complex kinetic systems. Apart from urging the interested reader to try writing his or her own 1D PIC code, I also advise him or her to read *Plasma Physics Via Computer Simulation* by Birdsall and Langdon [1]. Other books to look at include [2] and [3].

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